Probability Foundations

1. Random Experiment

Definition: A random experiment is an experiment whose outcome cannot be predicted with certainty before performing it.

Telugu: Oka experiment perform chese mundu exact result manaki teliyadu. Kaani possible results telustayi.

Examples:

- Tossing a coin.
- Rolling a die.
- Drawing a card.
- Selecting a student randomly.

Example 1: Tossing a coin

Possible outcomes:

- Head (H)
- Tail (T)

We cannot predict which one will occur.

Example 2: Rolling a die

Possible outcomes: 1, 2, 3, 4, 5, 6.

2. Sample Space (S)

Definition: The set of all possible outcomes of a random experiment is called the sample space.

Telugu: Oka experiment lo ravachu anukune anni possible outcomes collection ni Sample Space antaru.

Examples:

Experiment 1: Tossing one coin

$$S = \{H, T\}$$

Experiment 2: Rolling one die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Experiment 3: Tossing two coins

$$S = \{HH, HT, TH, TT\}$$

If a coin is tossed n times, the number of possible outcomes is:

$$|S| = 2^n$$

If a die is rolled n times:

$$|S| = 6^n$$

For two coins: $2^2 = 4$ outcomes.

3. Events

Definition: An event is any subset of the sample space.

Telugu: Sample space lo unna outcomes lo konni outcomes select chesukunte, adi Event.

Example: Rolling a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let A = Getting an even number.

$$A = \{2, 4, 6\}$$

Let B = Getting a number greater than 4.

$$B = \{5, 6\}$$

Types of events

Event	Meaning	Example
Simple event	One outcome	3
Compound event	Multiple outcomes	246
Certain event	Always occurs	S
Impossible event	Never occurs	∅
Complementary event	Event does not occur	A'
Mutually exclusive	Cannot occur together	A ∩ B = ∅

Important: If A is an event, its complement is:

$$A' = S - A$$

And:

$$P(A) + P(A') = 1$$

4. Event Operations

Consider:

$$A = \{2, 4, 6\}$$

$$B = \{4, 5, 6\}$$

Union (A ∪ B): A or B or both.

$$A \cup B = \{2, 4, 5, 6\}$$

Intersection (A ∩ B): Common elements.

$$A \cap B = \{4, 6\}$$

Difference (A − B): Elements in A but not in B.

$$A - B = \{2\}$$

These are very important because GATE questions frequently combine sets and probability.

5. Basic Probability Formula

When all outcomes are equally likely:

$$P(A) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$$

Example: Rolling a fair die. Find the probability of getting an even number.

Total outcomes = 6

Favourable outcomes = 6 = 3

$$P(A) = \frac{3}{6} = \boxed{\frac{1}{2}}$$

Important properties

1. $0 \leq P(A) \leq 1$
2. $P(S) = 1$
3. $P(\emptyset) = 0$
4. $P(A') = 1 - P(A)$
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

MCQ Test – 5 Questions

Q1. A coin is tossed 3 times. How many possible outcomes are there?

- A. 4
- B. 6
- C. 8
- D. 9

Q2. A die is rolled once. Event A = getting an odd number. What is A?

- A. {2,4,6}
- B. {1,3,5}
- C. {1,2,3}
- D. {5,6}

Q3. If P(A) = 0.35, find P(A').

- A. 0.35
- B. 0.65
- C. 1.35
- D. 0.50

Q4. Which event is impossible when rolling a standard die?

- A. Getting 6
- B. Getting an even number
- C. Getting 7
- D. Getting an odd number

Q5. If $A \cap B = \emptyset$, then A and B are:

- A. Independent events
- B. Mutually exclusive events
- C. Certain events
- D. Complementary in all cases

A fair coin is tossed 3 times. Which statements are correct?

- The probability of 3 heads is 3/8.
- The probability of exactly 2 heads is 3/8.
- The probability of at least 1 head is 7/8.
- The sample space contains 8 outcomes.

A fair six-sided die is rolled once. Let $A = \{2,4,6\}$ and $B = \{4,5,6\}$.
Which statements are correct?

- A and B are mutually exclusive.
- $A \cup B = \{2,4,5,6\}$
- $A \cap B = \{4,6\}$
- $A - B = \{2\}$

For any two events A and B in a probability space, which statements are always true?

- $P(A') = 1 - P(A)$
- $P(A \cup B) \leq 1$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cap B) = P(A)P(B)$

A standard fair die is rolled once. Which events have probability 1/2?

- Getting a number greater than 3
- Getting a prime number
- Getting an even number
- Getting a number less than 3

Two events A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.4$. Which statements are correct?

- $P(A \mid B) = 0.3$
- $P(A \cap B) = 0$
- $P(A \cup B) = 0.7$
- $P(A' \cap B) = 0.4$

A fair coin is tossed twice. Let A = {at least one head} and B = {exactly one head}. Which statements are correct?

- $P(A \cap B) = 1/2$
- A and B are mutually exclusive.
- $P(A) = 3/4$
- $P(B) = 1/2$

Which statements about probability are always correct?

- $P(\varnothing) = 0$
- $P(A)$ can be negative for some events.
- If $A \subseteq B$, then $P(A) \leq P(B)$.
- $P(S) = 1$

A card is drawn uniformly at random from a standard 52-card deck. Which statements are correct?

- The probability of drawing a king is 1/13.
- The probability of drawing a red card is 1/2.
- The probability of drawing a heart is 1/4.
- The probability of drawing a face card is 1/4.

Three fair coins are tossed simultaneously. Let X be the numbe of heads. Select every correct statement.

- $P(X \leq 1) = 5/8$
- $P(X = 1) = 3/8$
- $P(X \geq 2) = 1/2$
- $P(X = 0) = 1/8$

For any two events A and B in a probability space, which statements are always true, without assuming independence or mutual exclusivity?

- $P(A \cap B) = P(A)P(B)$
- $0 \leq P(A \cap B) \leq P(A)$
- $P(A \cup B) = P(A) + P(B)$
- $P(A \cup B) \leq 1$

Events A and B are mutually exclusive, with $P(A) = 0.25$ and $P(B) = 0.40$. Select every correct statement.

- $P(A \cup B) = 0.65$
- $P(A \cap B) = 0$
- A and B are independent
- $P(A' \cap B) = 0.40$

One card is drawn uniformly at random from a standard 52-card deck. Select every correct probability.

- P(a face card) = 1/4
- P(a queen) = 1/13
- P(a black card) = 1/2
- P(an ace or a king) = 2/13

A fair coin is tossed 4 times. Select every correct statement.

- [ ] Probability of no heads is 1/8
- [ ] Total number of possible outcomes is 16
- [ ] Probability of at least 1 tail is 15/16
- [ ] Probability of exactly 3 heads is 1/4

A probability function P is defined on a sample space S. Select every statement that follows from the probability axioms and their standard consequences.

- [ ] $P(A) + P(A') = 0$
- [ ] $P(A \cup A') = 0$
- [ ] If $A \subseteq B$, then $P(A) \leq P(B)$
- [ ] $P(A \cap A') = 0$

Events A and B satisfy $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Select every correct statement.

- [ ] $P(A' \cap B) = 0.2$
- [ ] A and B are independent
- [ ] $P(A \cup B) = 0.7$
- [ ] A and B are mutually exclusive

Q1. A fair six-sided die is rolled 3 times. Select every correct statement.

A The total number of ordered outcomes is 216.

B The probability of getting exactly two 6s is 5/72.

C The probability of getting no 6 is 125/216.

D The probability of getting at least one 6 is 1/6.

Q2. For events A and B, P(A) = 0.62 and P(B) = 0.28. Select every statement that must be true.

A $P(A') = 0.38$.

B $P(A \cap A') = 0$.

C $P(A \cup A') = 0.62$.

D $P(A \cup B)$ cannot exceed 0.90.

Q3. Events A and B have P(A) = 0.3, P(B) = 0.6 and P(A ∩ B) = 0.18. Select every correct statement.

A A and B are independent.

B A and B are mutually exclusive.

C $P(A \cup B) = 0.72$.

D $P(A' \cap B) = 0.42$.

Q4. Two cards are drawn simultaneously without replacement from a standard 52-card deck. Select every correct statement.

A The probability that both cards are kings is 1/221.

B The probability that both cards are from the same suit is 4/17.

C The probability that both cards are red is 25/102.

D The probability that one card is an ace and the other is a 2 is 8/663.

Quick formula revision

Concept	Formula
Complement	$1 - P(A)$
Union	$P(A) + P(B) - P(A \cap B)$
Intersection of independent events	$P(A)P(B)$
Mutually exclusive	$P(A \cap B) = 0$
Conditional probability	$P(A \mid B) = P(A \cap B)/P(B)$
At least one	$1 - P(\text{none})$
Union upper bound	$P(A \cup B) \leq P(A) + P(B)$
Union lower bound	$P(A \cup B) \geq \max(P(A), P(B))$

Q1. If P(A) = 0.35, what is P(A')?

- A. 0.35
- B. 0.65
- C. 1
- D. 0

Q2. If P(A) = 0.4 and P(B) = 0.3, what is the maximum possible value of P(A ∪ B)?

- A. 0.4
- B. 0.3
- C. 0.7
- D. 1

Q3. If two events are mutually exclusive, what is P(A ∩ B)?

- A. 1
- B. P(A) × P(B) always
- C. 0
- D. P(A) + P(B)

Q4. A card is drawn from a deck. What is the probability of getting a queen?

- A. 1/4
- B. 1/13
- C. 4/13
- D. 1/52

1. Conditional Probability

Meaning: Probability of an event happening when we already know that another event has happened.

A box contains:

- 3 red balls
- 2 blue balls

Total = 5 balls.

One ball is selected randomly.

Let:

- A = selecting a red ball.
- B = selecting a ball that is not blue.

What is the probability of A given B?

Since B means the selected ball is not blue, we already know that the ball is red.

Therefore:

$$P(A|B) = 1$$

Here, $P(A|B)$ means probability of A given B.

2. Conditional Probability Formula

This is an important GATE DA formula.

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

Condition: $P(B) > 0$.

Where:

- $P(A|B)$: Probability of A given B.
- $P(A \cap B)$: Probability that both A and B happen.
- $P(B)$: Probability that B happens.

3. Independent Events

Two events are independent if the occurrence of one event does not change the probability of the other event.

Telugu lo: Event A jarigina, Event B jarige probability change avvakapothe, avi independent events.

Formula

$$\boxed{P(A \cap B) = P(A) \times P(B)}$$

Equivalent condition:

$$\boxed{P(A|B) = P(A)}$$

4. Mutually Exclusive vs Independent

Very important GATE concept

Mutually exclusive	Independent
Cannot happen together	Can happen together
$P(A \cap B) = 0$	$P(A \cap B) = P(A)P(B)$
One happening prevents the other	One does not affect the other

Important: Two mutually exclusive events with positive probabilities cannot be independent.

Example:

- A = getting 2 on a die.
- B = getting 5 on the same die.

They are mutually exclusive because both cannot occur in one roll.

But they are not independent.

5. Multiplication Theorem

From conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Rearranging:

$$\boxed{P(A \cap B) = P(B) \times P(A|B)}$$

Similarly:

$$\boxed{P(A \cap B) = P(A) \times P(B|A)}$$

This is useful for questions involving drawing cards or balls without replacement.

6. Quick Revision Notes

Remember these five formulas:

Concept	Formula
Conditional probability	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Independent events	$P(A \cap B) = P(A)P(B)$
Multiplication theorem	$P(A \cap B) = P(A)P(B \mid A)$
Mutually exclusive	$P(A \cap B) = 0$
Conditional independence test	$P(A \mid B) = P(A)$

Q1. A class has 30 students. 18 know Python, 12 know SQL, and 8 know both. A student is selected randomly. Which statements are correct?

Select all correct options

- A. P(Python | SQL) = 2/3
- B. P(SQL | Python) = 4/9
- C. P(Python ∩ SQL) = 8/30
- D. P(Python | SQL) = 8/30

Q2. A fair die is rolled once. Let A = getting an even number and B = getting a number greater than 3. Select the correct statements.

Select all correct options

- A. A and B are independent
- B. P(A ∩ B) = 1/3
- C. P(A | B) = 2/3
- D. A and B are mutually exclusive

Q3. A bag contains 5 red and 3 blue balls. Two balls are drawn without replacement. Which statements are correct?

Select all correct options

- A. P(both red) = 5/14
- B. P(both blue) = 3/28
- C. P(first red, second blue) = 15/56
- D. P(second red | first red) = 4/7

Q4. Events A and B have P(A) = 0.4 and P(B) = 0.5. Which statements must be true if A and B are independent?

Select all correct options

- A. P(A ∩ B) = 0.2
- B. P(A | B) = 0.4
- C. P(B | A) = 0.5
- D. P(A ∪ B) = 0.9

Q1. Conditional Probability

A college has 40 students:

- 24 study Mathematics.
- 20 study Statistics.
- 12 study both.

Which statements are correct?

A. $P(M|S) = 3/5$

B. $P(S|M) = 1/2$

C. $P(M \cap S) = 3/10$

D. $P(M|S) = 12/40$

Q2. Without Replacement

A bag contains 6 red balls and 4 blue balls. Two balls are drawn without replacement.

Which statements are correct?

A. Probability of both red = $1/3$

B. Probability of both blue = $2/15$

C. Probability of first red and second blue = $4/15$

D. Probability of second red given first red = $5/9$

Q3. Conditional Probability

Two fair dice are rolled. Given that their sum is 8, which statements are correct?

A. Probability that the first die shows 3 is $1/5$.

B. Probability that the second die shows 5 is $2/5$.

C. Probability that both dice show the same number is $1/5$.

D. Probability that the first die shows 4 is $1/5$.

1. Addition Rule of Probability

The addition rule is used when a question asks for:

- A or B
- At least one event
- Either event happening

The basic formula is:

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

Here:

- $P(A)$: Probability of A.
- $P(B)$: Probability of B.
- $P(A \cap B)$: Probability of both A and B.

2. Special Case: Mutually Exclusive Events

If two events cannot happen together:

$$P(A \cap B) = 0$$

Therefore:

$$\boxed{P(A \cup B) = P(A) + P(B)}$$

3. Multiplication Rule of Probability

Used when the question asks for:

- A and B
- Both events happening
- Consecutive selections

Formula:

$$\boxed{P(A \cap B) = P(A) \times P(B|A)}$$

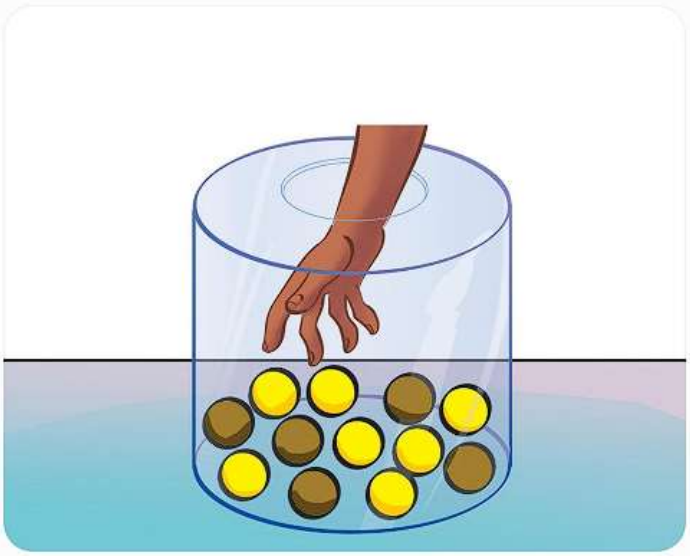
For independent events:

$$\boxed{P(A \cap B) = P(A) \times P(B)}$$

4. With Replacement vs Without Replacement

This is especially important for GATE DA.





With replacement

- Put the first ball back.
- Total remains unchanged.
- Draws are independent (for random draws).

Without replacement

- Do not put the ball back.
- Total decreases.
- Draws are dependent.

Find the probability of selecting two red balls.

Case A: With replacement

$$P(RR) = \frac{5}{8} \times \frac{5}{8} = \boxed{\frac{25}{64}}$$

Case B: Without replacement

$$P(RR) = \frac{5}{8} \times \frac{4}{7} = \boxed{\frac{5}{14}}$$

5. Quick Formula Revision

Situation	Formula
A or B	$P(A) + P(B) - P(A \cap B)$
Mutually exclusive	$P(A) + P(B)$
A and B	$P(A)P(B\|A)$
Independent events	$P(A)P(B)$
Neither A nor B	$1 - P(A \cup B)$
At least one	$1 - P(\text{none})$

Q1. A card is drawn from a standard deck of 52 cards. Which statements are correct for getting a Queen or a Diamond?

Select all correct options.

- A. P(Queen) = 1/13
- B. P(Diamond) = 1/4
- C. P(Queen ∩ Diamond) = 1/52
- D. P(Queen ∪ Diamond) = 4/13

Q2. A fair die is rolled once. Let A = getting an odd number and B = getting a number greater than 4. Select the correct statements.

Select all correct options.

- A. P(A ∪ B) = 5/6
- B. P(A ∩ B) = 1/6
- C. A and B are mutually exclusive
- D. P(A ∪ B) = 1

Q3. A bag contains 4 red and 6 blue balls. Two balls are drawn with replacement. Which statements are correct?

Select all correct options.

- A. P(both red) = 4/25
- B. P(both blue) = 9/25
- C. P(first red, second blue) = 6/25
- D. The two draws are independent

Q4. Events A and B are independent, with P(A) = 0.3 and P(B) = 0.6. Which statements are correct?

Select all correct options.

- A. P(A ∩ B) = 0.18
- B. P(A ∪ B) = 0.72
- C. P(neither A nor B) = 0.28
- D. P(A | B) = 0.3

1. Law of Total Probability

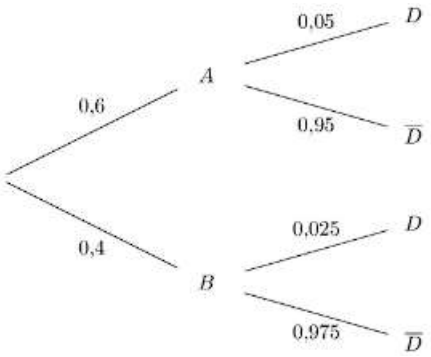
First, understand this idea:

Suppose a company manufactures products using two machines, Machine A and Machine B.

- Machine A produces 60% of products.
- Machine B produces 40% of products.
- Some products from each machine are defective.

If we randomly select one product, what is the probability that it is defective?

We need to consider both possible sources of the product.



Formula

If $B_1, B_2, \ldots, B_n$ are mutually exclusive and collectively exhaustive events, then:

$$\boxed{P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)}$$

For two machines:

$$\boxed{P(D) = P(D|A)P(A) + P(D|B)P(B)}$$

Where:

- D = defective product.
- A, B = product manufactured by Machine A or B.

2. Bayes' Theorem

Now let's make the problem more interesting.

Suppose we randomly select a defective product.

What is the probability that it came from Machine B?

Notice the difference:

- Total probability: Find the probability that the product is defective.
- Bayes' theorem: Given that the product is defective, find its source.

This reverse calculation is called Bayes' theorem.

Bayes' Theorem Formula

$$\boxed{P(B|A) = \frac{P(A|B)P(B)}{P(A)}}$$

3. Prior Probability vs Posterior Probability

These two terms are important.

Term	Meaning	Example
Prior	Probability before new evidence	$P(B) = 0.4$
Likelihood	Probability of evidence given a hypothesis	$P(D|B) = 0.05$
Evidence	Probability of observing the evidence	$P(D) = 0.032$
Posterior	Updated probability after evidence	$P(B|D) = 0.625$

Remember:

- Prior = Before evidence.
- Posterior = After evidence.

1. Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

2. Total probability

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

3. Bayes' theorem

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$$

4. Complement rule

$$P(\bar{A}) = 1 - P(A)$$

Q1. Two machines A and B produce 70% and 30% of products respectively. Their defective rates are 1% and 4%. Which statements are correct?

Select all correct options.

- A. P(D) = 0.019
- B. P(B | D) = 12/19
- C. P(A | D) = 7/19
- D. P(D | A) = 0.7

Q2. A box is selected randomly: Box 1 with probability 0.6 or Box 2 with probability 0.4. Box 1 contains 50% red balls, while Box 2 contains 25% red balls. Which statements are correct?

Select all correct options.

- A. P(R) = 0.4
- B. P(Box 1 | R) = 0.75
- C. P(Box 2 | R) = 0.25
- D. P(R | Box 1) = 0.6

Q3. Three machines produce 20%, 30%, and 50% of products respectively. Their defective rates are 1%, 2%, and 4%. Which statements are correct?

Select all correct options.

- A. Overall defective probability = 0.028
- B. P(Machine 3 | D) = 5/7
- C. P(Machine 1 | D) = 1/14
- D. Overall defective percentage = 2.8%

Q4. A disease affects 2% of a population. A test has 90% sensitivity and 95% specificity. Which statements are correct?

Select all correct options.

- A. P(positive test) = 0.067
- B. P(disease | positive) = 18/67
- C. False positive rate = 0.05
- D. P(disease | positive) = 0.9

Q5. Events B1 and B2 form a partition of the sample space. Given P(B1) = 0.4, P(B2) = 0.6, P(A | B1) = 0.2, and P(A | B2) = 0.5, which statements are correct?

Select all correct options.

- A. P(A) = 0.38
- B. P(B1 | A) = 4/19
- C. P(B2 | A) = 15/19
- D. P(A | B2) = 0.38

Section A – Mixed MSQs

Q1. Probability and Set Theory (MSQ)

Let A and B be two events such that:

$$P(A) = 0.65, \quad P(B) = 0.55$$

Which statements must be true?

A. $P(A \cup B) \geq 0.65$

B. $P(A \cap B) \geq 0.20$

C. $P(A \cap B) \leq 0.55$

D. $P(A^c \cup B^c) = 1 - P(A \cap B)$

Q2. Independent vs Mutually Exclusive (MSQ)

Two events A and B satisfy:

$$P(A) = 0.4, \quad P(B) = 0.3$$

Which statements are correct?

A. If A and B are independent, $P(A \cap B) = 0.12$.

B. If A and B are mutually exclusive, $P(A \cup B) = 0.7$.

C. A and B cannot be both independent and mutually exclusive.

D. If A and B are independent, $P(A \cup B) = 0.58$.

Q3. Conditional Probability (MSQ)

A bag contains 5 red, 4 green and 3 yellow balls. Two balls are drawn without replacement.

Which statements are correct?

A. Probability that both are red = $5/33$.

B. Probability that both are of the same colour = $19/66$.

C. Probability that the second ball is red given the first is green = $5/11$.

D. Probability that at least one ball is yellow = $17/44$.

Q4. Cards (MSQ)

One card is drawn randomly from a standard 52-card deck.

Which statements are correct?

A. Probability of drawing a face card = $3/13$.

B. Probability of drawing a red King = $1/26$.

C. Probability of drawing a Heart or a face card = $11/26$.

D. Given that the card is red, probability that it is a Queen = $1/13$.

Q5. Total Probability and Bayes' Theorem (MSQ)

A company uses three suppliers:

Supplier	Production share	Defective rate
A	50%	2%
B	30%	3%
C	20%	5%

A randomly selected product is found to be defective.

Which statements are correct?

A. Overall defective probability = 0.029.

B. Probability that the defective product came from supplier A = $10/29$.

C. Probability that it came from supplier C = $10/29$.

D. Probability that it came from supplier B = $9/29$.

Q6. Dice and Conditional Probability (MSQ)

Two fair dice are rolled. Given that their sum is at least 10, which statements are correct?

A. Probability that both dice show the same number = $1/2$.

B. Probability that the first die shows 6 = $1/2$.

C. Probability that the sum is exactly 12 = $1/6$.

D. Probability that at least one die shows 6 = $5/6$.

Q7. Mixed Probability (MSQ)

Events A and B satisfy:

$$P(A) = 0.5, \quad P(B) = 0.4, \quad P(A \cup B) = 0.7$$

Which statements are correct?

A. $P(A \cap B) = 0.2$.

B. A and B are independent.

C. $P(A|B) = 0.5$.

D. $P(A^c \cap B^c) = 0.3$.

Section B – MCQs

Q8. Bayes' Theorem (MCQ)

There are two boxes:

- Box 1 is selected with probability $1/3$.
- Box 2 is selected with probability $2/3$.

Box 1 contains 3 red and 2 blue balls.

Box 2 contains 2 red and 4 blue balls.

One ball is drawn from the selected box and is found to be red.

What is the probability that Box 1 was selected?

A. $3/7$

B. $5/11$

C. $9/17$

D. $2/5$

Q9. Without Replacement (MCQ)

A bag contains 7 white and 5 black balls. Three balls are selected without replacement.

What is the probability that exactly two balls are white?

A. $7/22$

B. $21/44$

C. $35/66$

D. $5/11$

Q10. Conditional Independence (MCQ)

Events A and B satisfy:

$$P(A) = 0.6, \quad P(B) = 0.5$$

$$P(A \cap B) = 0.3$$

Which statement is correct?

A. A and B are mutually exclusive.

B. A and B are independent.

C. A and B are dependent.

D. $P(A|B) = 0.5$.